Course 2
Algorithm Representation

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Algoritma Pemrograman (IF201)
Institut Teknologi Tangerang Selatan
Semester Ganjil 2026/2027

Objectives

- To describe a simple algorithm in clear, ordered steps.
- To express the same solution using pseudocode and a flowchart.
- To use common flowchart symbols correctly.
- To trace an algorithm and check its result with sample data.

SUB-CPMK-IF201-1-2: Explain the basic principles of algorithms and programming.

What Do You Remember?

- What makes a sequence of steps an algorithm?
- What are the input, process, and output of a shopping calculation?
- How is an algorithm different from a program?
- Could two people describe the same algorithm differently?

Why Represent an Algorithm?

- We need a way to explain our solution before writing code.
- A clear representation helps another person follow the logic.
- It helps us identify missing steps and unclear assumptions.
- It gives us something to check with sample input values.

Could someone implement your solution from your explanation?

Three Ways to Represent an Algorithm

Representation	How we describe the solution	What to watch for
Narrative	Ordered steps in everyday language	Ambiguous or missing steps
Pseudocode	Structured instructions in text	Inconsistent notation or order
Flowchart	Symbols connected by flow lines	Incorrect symbols or paths

What should remain the same across all three representations?

Our Case: A Simple Shopping Bill

A customer buys several items at the same price.

Part	Description
Input	price: price per item; quantity: number of items
Process	Multiply price by quantity
Output	total: amount to pay

Assume price ≥ 0 and quantity is a positive whole number.
This case has no discount, tax, or extra charge.

Narrative Description

- Start.
- Read the price per item and the quantity.
- Multiply the price by the quantity to obtain the total.
- Display the total amount to pay.
- Stop.

Could you carry out these steps without writing a program?

Is This Description Clear Enough?

"Take the shopping data, calculate the bill, and display the correct amount."

- Which values do we need from the customer?
- What calculation produces the bill?
- What assumptions should we state?
- How would you rewrite this as ordered steps?

What is Pseudocode?

- Pseudocode expresses an algorithm using structured, readable text.
- It focuses on the order of operations and the meaning of each step.
- It does not require the exact syntax of a programming language.
- Different conventions are possible. Use one convention consistently.

Would changing the wording always change the algorithm?

Our Pseudocode Convention

Instruction	Meaning	Example
START / END	Beginning and end	START
INPUT	Read a value	INPUT price, quantity
SET	Store a calculated value	SET total = price * quantity
OUTPUT	Display a result	OUTPUT total
IF / ELSE / ENDIF	Choose a path from a condition	IF total >= 100000 THEN

In SET, “=” means assign a value. In a condition, “>=” means greater than or equal to.

Shopping Bill: Pseudocode

```
START
    INPUT price, quantity
    SET total = price * quantity
    OUTPUT total
END
```

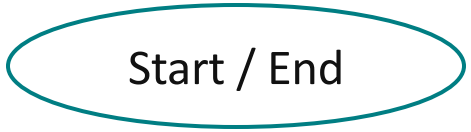
Why must we read price and quantity before calculating total?

A Dry Run: Following the Steps

Step	price	quantity	total	Visible output
INPUT values	5000	3	Not set	None
SET total	5000	3	15000	None
OUTPUT total	5000	3	15000	15000

A dry run means following the instructions manually and recording what changes at each step.

Common Flowchart Symbols



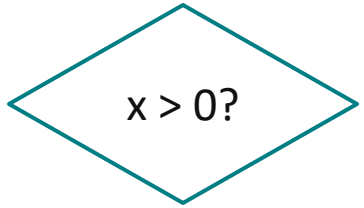
Terminator: where the algorithm begins or ends.



Input/output: reading data or showing a result.



Process: a calculation or an assignment.



Decision: a condition with labeled outcomes.



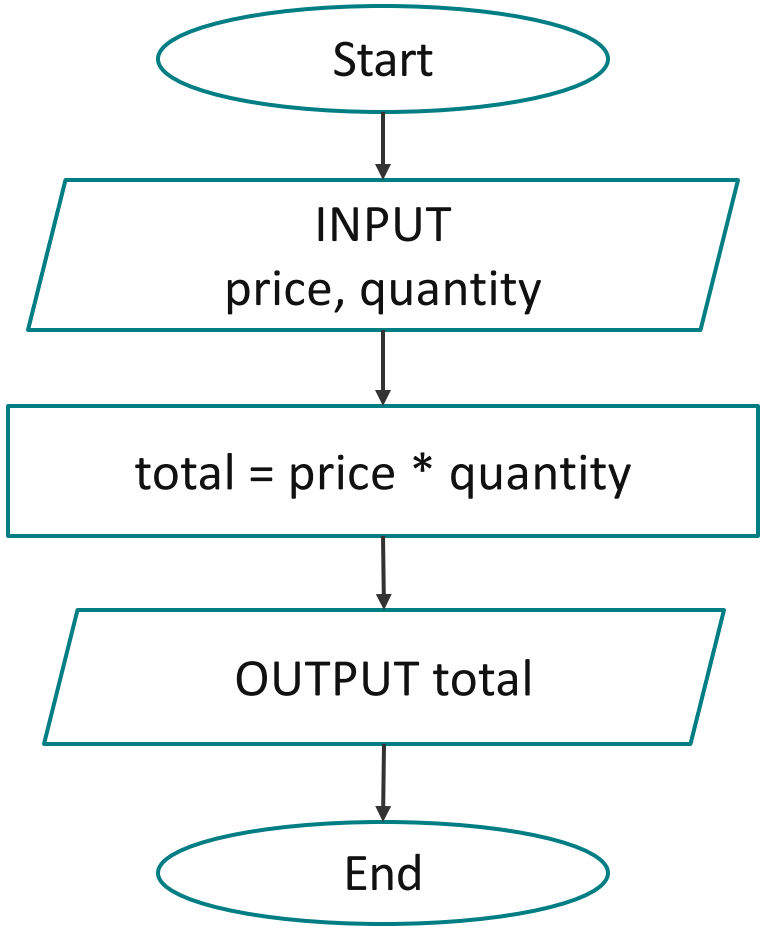
Flow line: an arrow that shows the next step.

What Makes a Flowchart Readable?

- Use a clear start point and an end point.
- Follow a consistent direction, usually from top to bottom.
- Choose the symbol that matches the operation.
- Label decision paths, for example Yes and No.
- Check that each path leads to the intended next step.

What could happen if one decision path has no destination?

Shopping Bill: Flowchart



- Read the input.
- Calculate the total.
- Display the result.
- Follow the same sequence as the pseudocode.

Where does the calculation happen in this diagram?

What if the Bill Includes a Discount?

- Suppose the shop gives a 10% discount when the total is at least Rp100,000.
- Otherwise, the customer pays the full total.
- Input: total before discount. Output: payable amount.
- Assume total is nonnegative. No other charges apply.

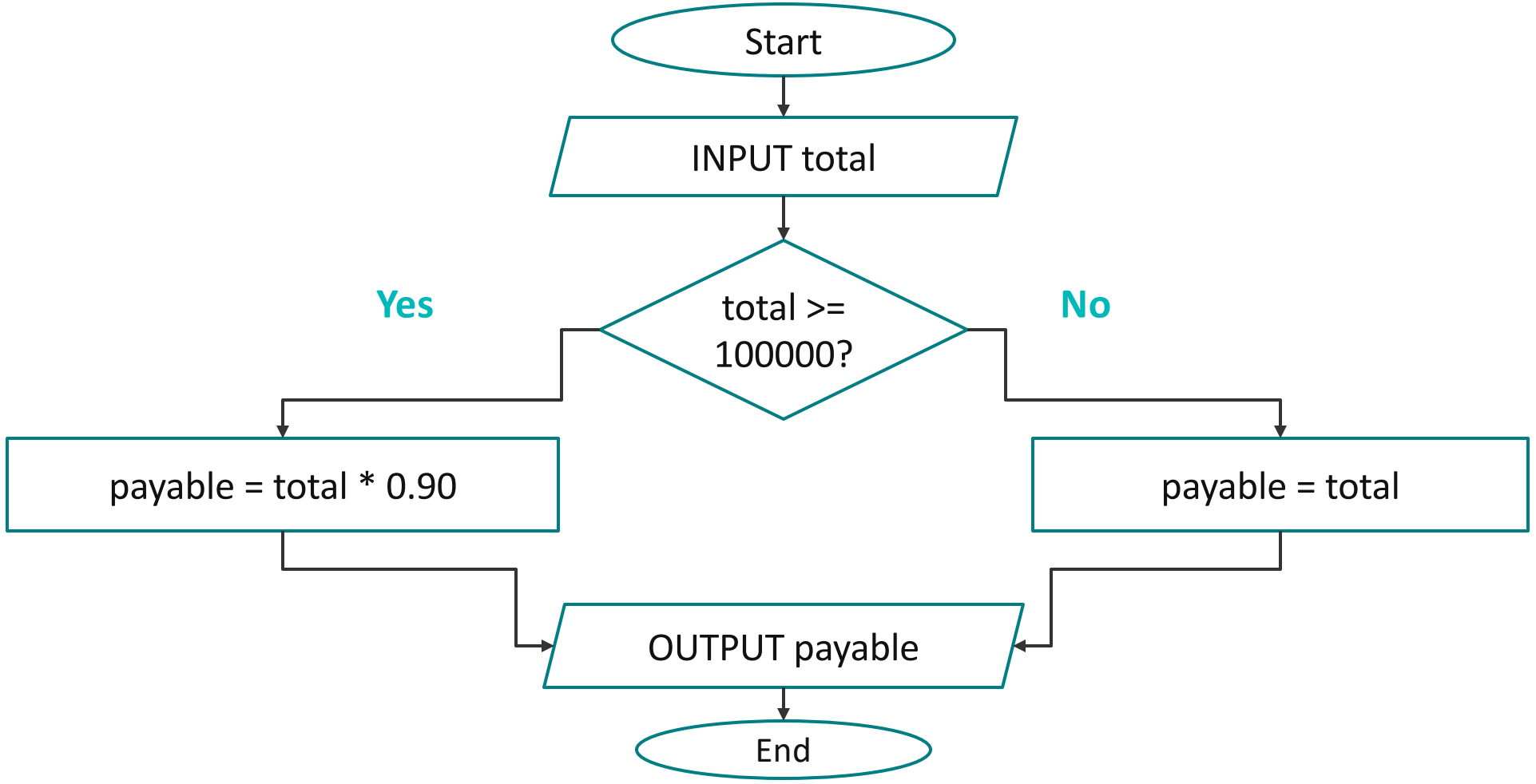
Does a total of exactly Rp100,000 qualify? Why?

A Decision in Pseudocode

```
START
    INPUT total
    IF total >= 100000 THEN
        SET payable = total * 0.90
    ELSE
        SET payable = total
    ENDIF
    OUTPUT payable
END
```

Why is OUTPUT payable placed after ENDIF?

A Decision in a Flowchart



Testing Both Paths

Input total	total >= 100000?	Expected payable	What it checks
Rp99,999	No	Rp99,999	Just below the threshold
Rp100,000	Yes	Rp90,000	Exactly at the threshold
Rp150,000	Yes	Rp135,000	Above the threshold

What would go wrong if the condition used “>” instead of “>=”?

Your Turn: Area of a Rectangle

- Input: length and width. Assume both are positive.
- Write an ordered narrative description.
- Write pseudocode for the same solution.
- Draw a flowchart with the appropriate symbols.
- Test length = 4 and width = 3. What should the output be?

Can your partner follow your solution without asking for extra steps?

Checking Each Other's Work

- Do the narrative, pseudocode, and flowchart describe the same logic?
- Does each variable have a value before a step uses it?
- Does every symbol match its operation?
- If there is a decision, are both paths labeled and complete?
- Does the sample output match the result you expected?

Quick Check

- Which symbol should represent reading a value?
- What is missing if we calculate total but never display it?
- Does changing a flowchart’s layout change its logic?
- How can a dry run reveal a mistake before we write code?

Which representation helps you explain your reasoning most clearly?

Closing Remarks

- How would you explain the same algorithm to a classmate and to a programmer?
- Which representation makes the order of steps easiest to follow?
- What did the dry run reveal about your own solution?
- A clear representation should communicate the logic and support testing.

Next topic: Basic Elements and Program Logic

Variables, constants, data types, operators, and control flow.

Thank You

Terima Kasih

Matur Kaseh

Hatur Nuhun

Kurre Sumanga

Matur Suwun

Tarimo Kasih

Tanki Dapa

Suksma

Wa…Wa

Nggahi Weki

Mbore

Nuhan

Mauliate

Tabea

Ami Ibi

Tinggilan

Ofo Nai

Ti Mopohula To

Kando Kami

Maita Komo

Kaseh Lima